UNIVERSITI TEKNOLOGI MARA
FINAL EXAMINATION

COURSE	PROGRAMMING I
COURSE CODE	CSC402
EXAMINATION	NOVEMBER 2026
TIME	3 HOURS

INSTRUCTIONS TO CANDIDATES

1. This question paper consists of the questions printed in this booklet.
2. Answer ALL questions in the Answer Booklet. Start each answer on a new page.
3. Do not bring any material into the examination room unless permission is given by the invigilator.
4. Please check to make sure that this examination pack consists of:
   i) the Question Paper
   ii) an Answer Booklet - provided by the Faculty
5. Answer ALL questions in English.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

This examination paper consists of 6 printed pages.

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PART A (10 MARKS)

Answer ALL questions.

1. Which declaration creates an integer array of five elements?
   A. int a(5);
   B. int a;
   C. int a[5];
   D. array a[5];

2. Given the declaration, what is displayed?

```
char word[6] = {'p', 'r', 'i', 'n', 't', '\0'};
cout << word[3];
```

   A. p
   B. t
   C. i
   D. n

3. Which identifier is valid in C++?
   A. total_score
   B. 2total
   C. total score
   D. int

4. What is displayed by the following code?

```
int x = 5;
cout << x * 2;
```

   A. 5
   B. 7
   C. 25
   D. 10

5. A computer system's hardware includes which of the following?
   i. input devices
   ii. output devices
   iii. central processing unit
   iv. main memory
   v. secondary storage
   A. i, iii, iv
   B. i, ii, iii, iv, v
   C. i, ii, iii, iv
   D. i, ii, iii, v

6. Which operator reads data from cin?
   A. %
   B. <<
   C. =
   D. >>

7. Which of the following are selection structures?
   i. if

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ii. switch
iii. for
iv. while
A. i and iii
B. iii and iv
C. i and ii
D. ii and iv

8. What does the % operator return?
A. The remainder of integer division
B. The quotient
C. The square root
D. The absolute value

9. Which loop is guaranteed to execute its body at least once?
A. do...while
B. while
C. if
D. for

10. Which structure chooses between two alternative paths?
A. struct
B. array
C. if...else
D. for

PART B (60 MARKS)

QUESTION 1

This question tests high-level languages and debugging.

a) Define syntax error and logic error.

(3 marks)

b) Explain TWO advantages of high-level languages over machine language.

(2 marks)

QUESTION 2

A concert charges the ticket prices shown in Table 1. A negative age is invalid.

Table 1. Concert Ticket Prices

Age Range	Ticket Price (RM)
0 - 12	15.00
13 - 17	25.00
18 and above	40.00

a) Draw a flowchart for the ticket price selection.

(5 marks)

b) Write pseudocode and a complete C++ program for the ticket price selection.

(8 marks)

QUESTION 3

This question tests declarations, constants and expressions.

a) Write C++ declarations for: an integer age, a double price, a constant discount rate of 0.10 and a character grade.

(6 marks)

b) Write C++ statements to read age and price, calculate price after discount, and display the result.

(5 marks)

QUESTION 4

Study the following array program segment:

```
int a[5] = {2, 4, 6, 8, 10};
int s = 0;
for (int i = 0; i < 5; i++) {
    s += a[i];
}
cout << s;
```

a) Trace the program and state the output.

(6 marks)

b) Rewrite the segment using a do...while loop that sums only values greater than 5.

(9 marks)

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QUESTION 5

A sports club records the sport names and medal counts in Table 1.

a) Declare two parallel arrays for the five sports in Table 1 and initialise them with those values.

Table 2. Sport and Medal Records

Sport	Medals
Swimming	8
Badminton	12
Athletics	7
Cycling	5
Hockey	9

(8 marks)

b) Write a program segment to display each sport with its medals, then calculate the total and average medals.

(8 marks)

PART C (30 MARKS)

QUESTION 1

A lecturer records the marks of ten students in an array.

Write a complete C++ program that reads ten marks, displays the average, the highest mark and its index, and counts how many marks are above the average.

(15 marks)

QUESTION 2

A cafe sells the items shown in Table 1. A 10% discount is given when the subtotal is at least RM50.

Table 3. Cafe Menu

Item	Price (RM)
Coffee	8.50
Tea	6.00
Juice	9.50

Write a complete C++ program that reads the item code and quantity, calculates the subtotal and discount, and displays a formatted receipt with the item name, quantity, subtotal, discount and amount to pay.

(15 marks)

END OF QUESTION PAPER

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